Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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Introduction:

The objective of this project was to develop a complete end-to-end data science solution using data collected from the Books to Scrape website. Books to Scrape is a practice website designed for web scraping and data analysis activities. The project focused on collecting book information, preparing the data for analysis, exploring relationships between variables, and developing a predictive model capable of estimating book prices based on available features.

The project demonstrates the practical application of web scraping, data cleaning, exploratory data analysis, data visualisation and machine learning techniques. These processes reflect the workflow commonly followed by data scientists when working with real-world business problems.

Report:

Data Acquisition

The dataset used in this project was collected directly from the Books to Scrape website. A web scraping solution was developed using Python and the BeautifulSoup library to extract data from multiple pages of the online catalogue.

The following information was collected for each book:

- Book Title
- Price (GBP)
- Star Rating
- Stock Availability

More than 200 books were collected to satisfy the project requirements. The extracted information was organised into a structured dataset and saved as a CSV file for further analysis.

The data acquisition process provided a reliable and reproducible method of gathering information from a publicly available online source.

Data Cleaning and Preparation

After the data collection stage, the dataset was inspected and cleaned to ensure that it was suitable for analysis and modelling.

The first step involved checking the dataset for missing values and duplicate records. Duplicate entries were removed where necessary to improve data quality.

The price variable was originally stored as a text field containing currency symbols. These symbols were removed and the values were converted into a numeric format to enable statistical analysis.

The rating values were initially represented as text categories ranging from One to Five. These categories were transformed into numerical values ranging from 1 to 5 to facilitate mathematical calculations and predictive modelling.

The cleaned dataset provided a structured and consistent foundation for exploratory analysis and machine learning.

Data Visualisation and Exploration

Several visualisations were created to explore relationships within the dataset and identify patterns in book pricing and ratings.

Price Distribution

Insert Histogram Screenshot Here

The histogram was used to examine the distribution of book prices. The visualisation showed how prices were spread across the dataset and highlighted the most common price ranges.

Rating Distribution

Insert Count Plot Screenshot Here

The rating distribution chart illustrated the frequency of books within each rating category. This chart provided insight into how books were generally rated within the dataset.

Price by Rating

Insert Box Plot Screenshot Here

The box plot compared book prices across rating categories. This visualisation was used to determine whether higher-rated books tended to have higher prices.

Price Versus Rating

Insert Scatter Plot Screenshot Here

The scatter plot was used to investigate the relationship between ratings and prices. The visualisation provided an indication of whether a relationship existed between these variables.

Correlation Heatmap

Insert Heatmap Screenshot Here

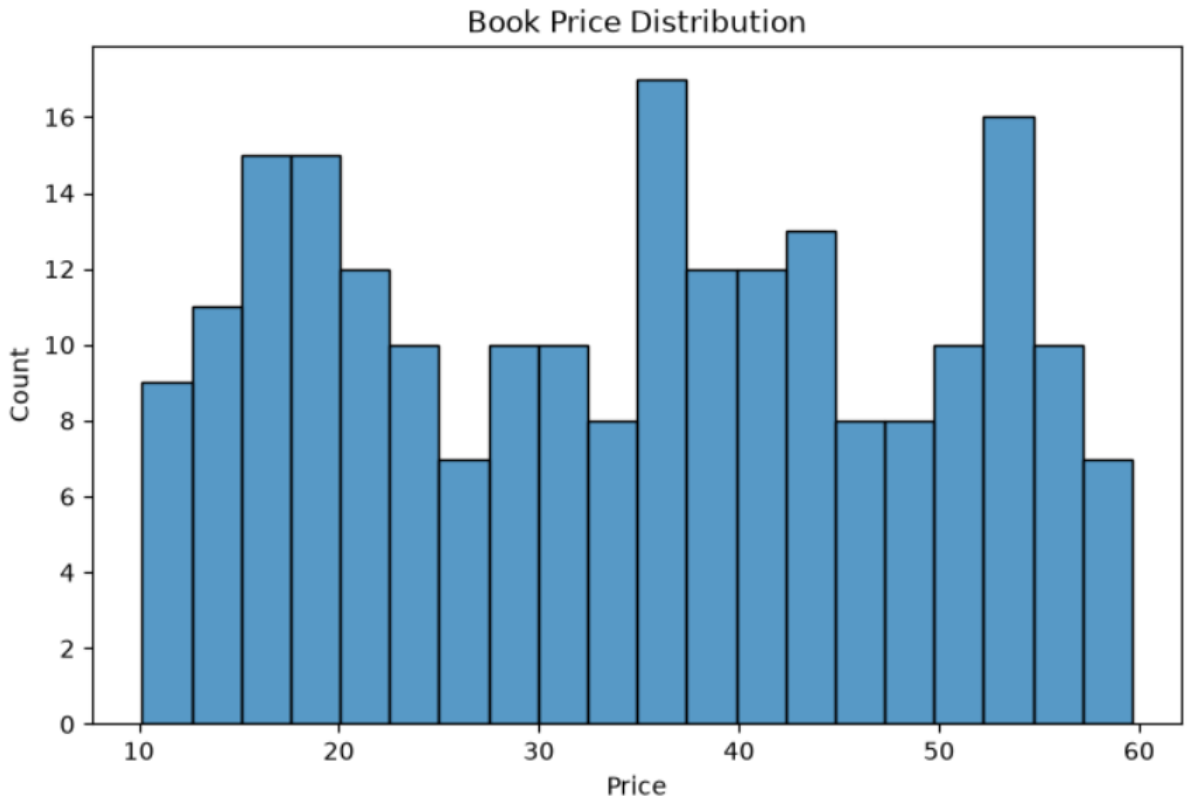
The heatmap displayed the correlation between numerical variables within the dataset. Correlation values provided insight into the strength and direction of relationships between variables.

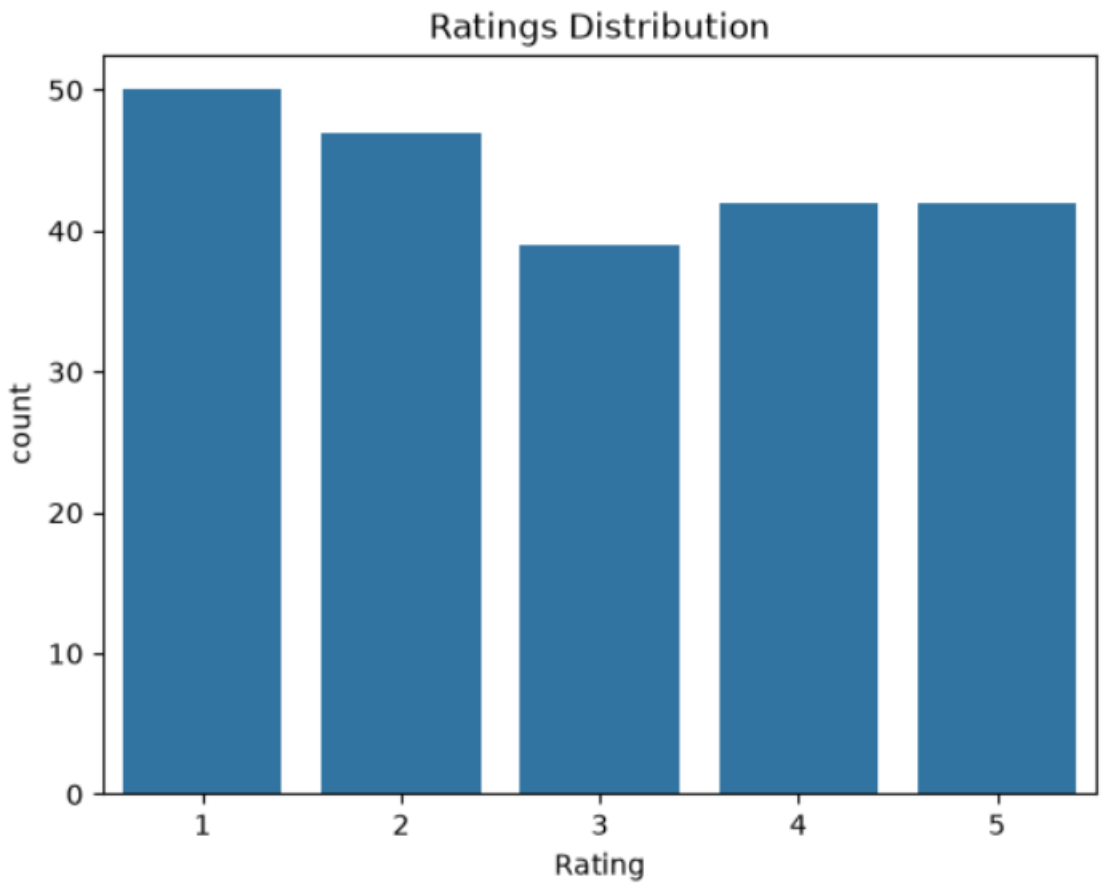
Top 10 Most Expensive Books

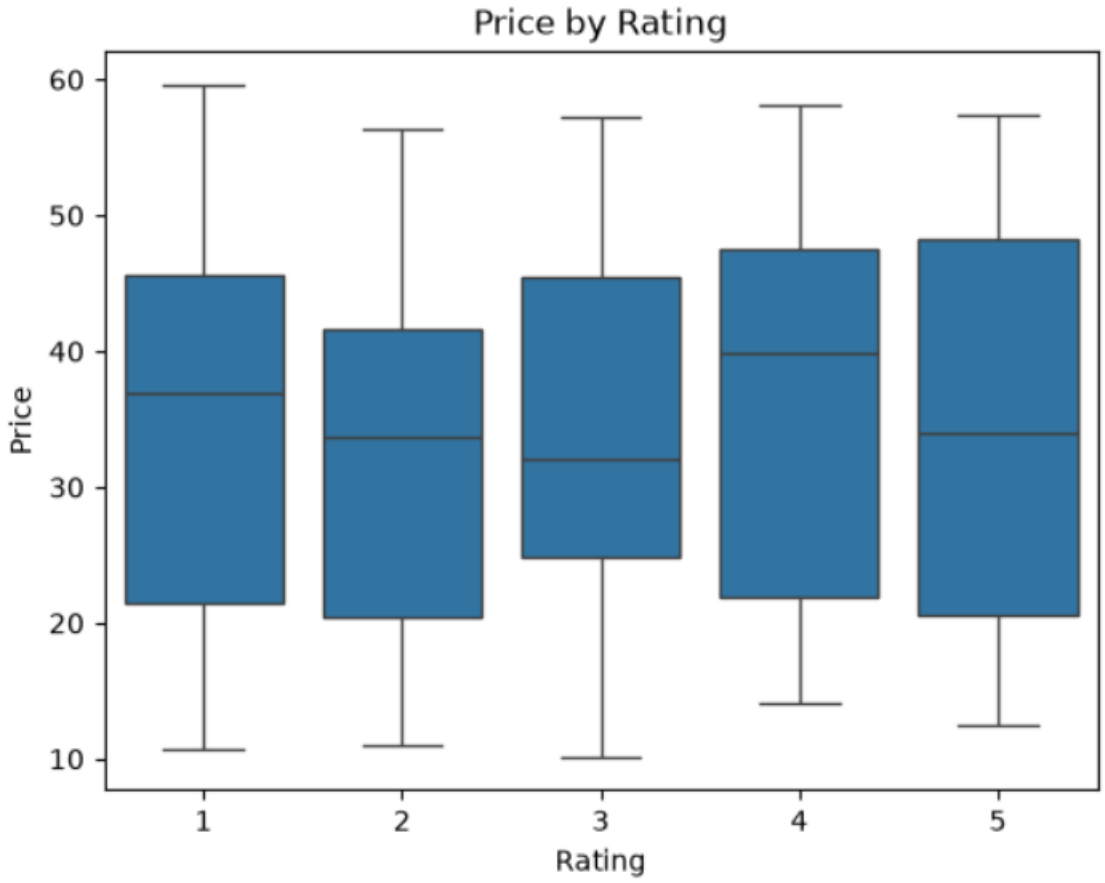
Insert Bar Chart Screenshot Here

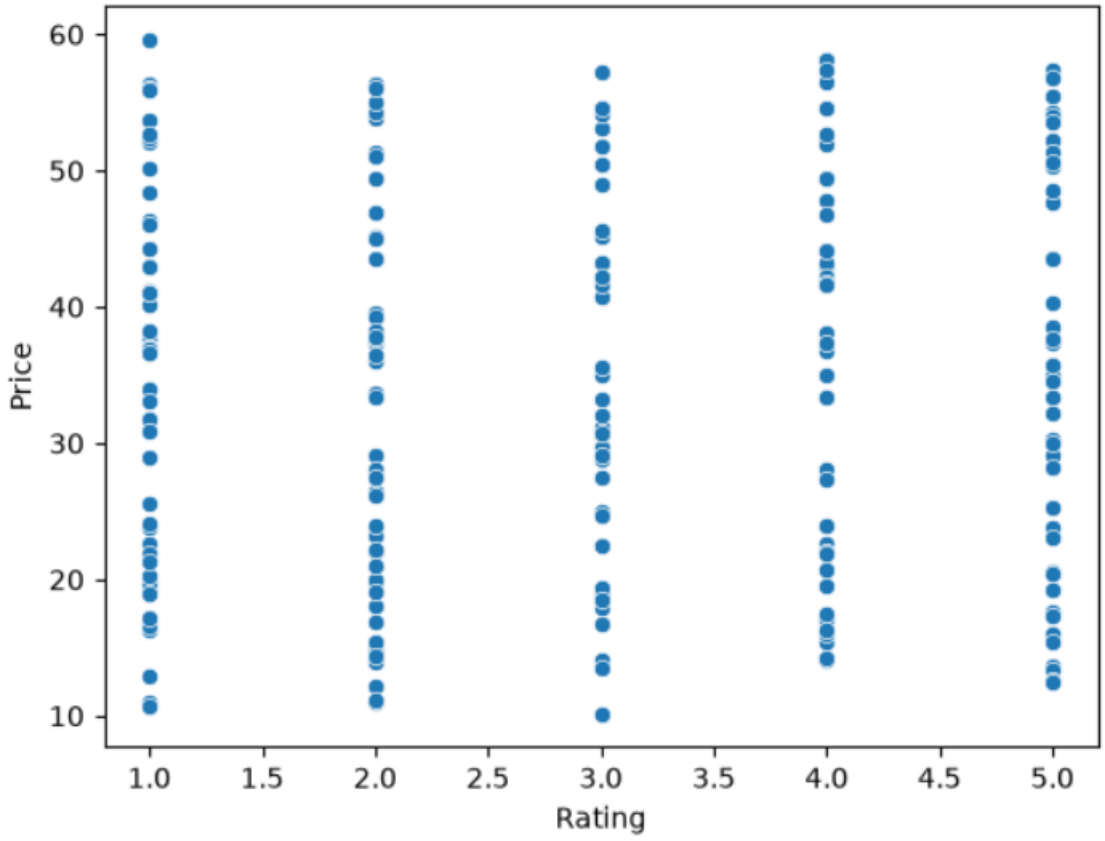
The bar chart highlighted the highest-priced books contained in the dataset and provided an overview of premium products available within the catalogue.

Visualisation graphs

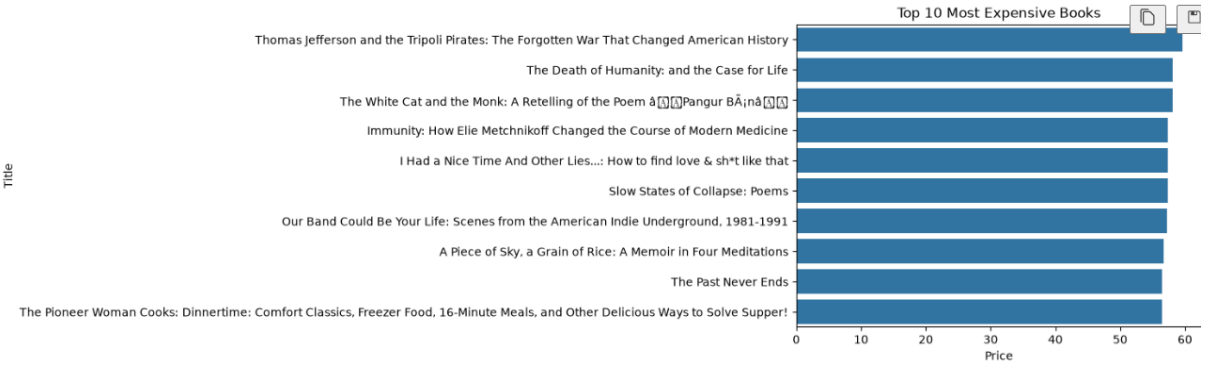



Ratings Distribution
count
0
10
20
30
40
50
1
2
3
4
5
Rating


Price by Rating
Price
Rating
60
50
40
30
20
10
1
2
3
4
5


Price
Rating
60
50
40
30
20
10
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Price
Rating
1
0.023
0.023
1
Price
Rating
1.0
0.8
0.6
0.4
0.2


Top 10 Most Expensive Books
Thomas Jefferson and the Tripoli Pirates: The Forgotten War That Changed American History
The Death of Humanity: and the Case for Life
The White Cat and the Monk: A Retelling of the Poem â□□Pangur BÃ¡nâ□□
Immunity: How Elie Metchnikoff Changed the Course of Modern Medicine
I Had a Nice Time And Other Lies...: How to find love & sh*t like that
Slow States of Collapse: Poems
Our Band Could Be Your Life: Scenes from the American Indie Underground, 1981-1991
A Piece of Sky, a Grain of Rice: A Memoir in Four Meditations
The Past Never Ends
The Pioneer Woman Cooks: Dinnertime: Comfort Classics, Freezer Food, 16-Minute Meals, and Other Delicious Ways to Solve Supper!
Title
0
10
20
30
40
50
60
Price

Predictive Modelling

A Linear Regression model was developed to predict book prices using available rating information.

The cleaned dataset was divided into training and testing subsets. The training dataset was used to build the model, while the testing dataset was used to evaluate predictive performance.

Linear Regression was selected because of its simplicity and suitability for examining relationships between variables. The model attempted to estimate book prices based on rating values and determine the extent to which ratings could explain price variations.

Model performance was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R^2 score.

The evaluation metrics provided insight into the accuracy of the model and its ability to make reliable predictions.

Key Findings

The analysis revealed several important findings.

Book prices displayed a broad distribution, indicating significant variability across the catalogue.

Ratings were concentrated within a limited range, suggesting that many books received similar evaluations.

Visualisations indicated that ratings alone had a relatively weak relationship with book prices. Books with similar ratings often displayed substantially different prices.

The predictive model demonstrated that ratings provided some information regarding price, but additional variables would likely be required to achieve highly accurate predictions.

The findings suggest that book pricing is influenced by multiple factors and cannot be fully explained by ratings alone.

Recommendations

Based on the analysis conducted, the following recommendations are proposed:

For Retailers

Retailers should consider multiple factors when developing pricing strategies rather than relying solely on customer ratings.

For Publishers

Publishers should investigate additional factors that influence pricing, such as category, author popularity and consumer demand.

For Future Data Science Projects

Future predictive models should incorporate additional features to improve prediction accuracy. Potential features include:

- Book Category
- Product Description
- Stock Quantity
- Text-Based Features
- Product Metadata

Including additional variables may significantly improve model performance and provide deeper business insights.

Conclusion:

This project successfully demonstrated the complete data science workflow from data acquisition to predictive modelling.

Data was collected from the Books to Scrape website, cleaned and prepared for analysis, visualised to gain insight into pricing trends, and used to develop a Linear Regression model.

The results showed that ratings alone provide limited explanatory power for predicting book prices. While the model was able to identify general relationships, additional features would be required for stronger predictive performance.

The project highlighted the importance of data cleaning, exploratory analysis and model evaluation when solving practical business problems using data science techniques.

GitHub Link:

thash2007/DS-500

Reference List:

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